WEATHER CLIMATE WATER
TEMPS CLIMAT EAU

DAYCLI message

Expert Team on Data Development and Stewardship (ET-DDS)

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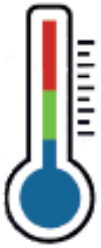
Task Team on Table Driven Code Forms (TT-TDCF) and Expert Team on Data Standards (ET-DATA)

WMO OMM

World Meteorological Organization
Organisation météorologique mondiale

DAYCLI message?

A WMO message with a worldwide mission to share daily values of temperature, precipitation and snow, currently at a month frequency.



Daily Minimum Mean and Maximum temperatures



Daily total accumulated precipitation



Daily total snow depth & Daily depth of fresh snow

Its special feature is the exchange of data and metadata on the quality of the measurements, including the **time period of each variable**, the **computation method**, the **siting classification**, the **measurement quality classification** and the **data quality information**, all of which contribute to better climate services.

It enables the **climatology departments** of each NMHS to assess and then share their **climate datasets** nationally, regionally and globally.
Therefore, it is important to respect the integrity of these datasets at the different levels and take into account the national climatological practices of each NMHS.

Which modifications?

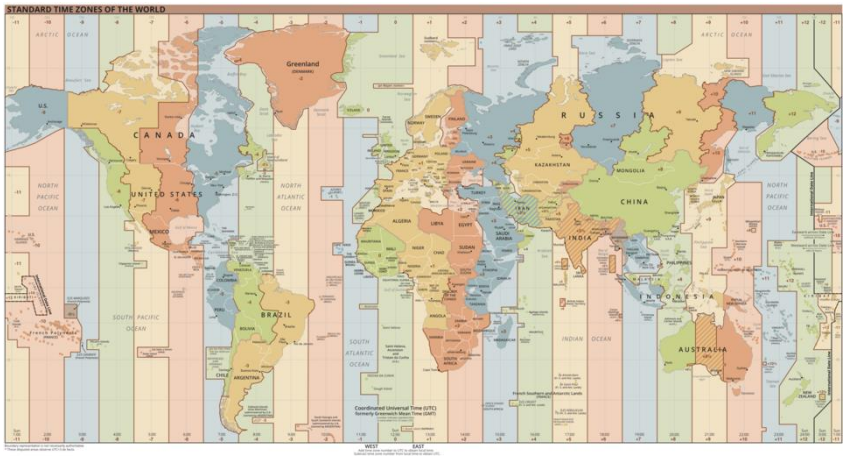
2 modifications:

- **Reference Date and Time (RDT)** used for each station by the NMHS to store its climate data **instead of UTC.** It represents the day (the 24 time period) on which the measurement values are attributed.

- Add the Time Difference (offset) between the **Reference Date and Time (RDT)** of the station measurement and **UTC**.

Note:

- The **RDT** used by the NMHS for a station to store its climate data does not necessarily correspond to the **standard time zone** in which the meteorological station is located.
- An NMHS may manage several **RDTs**.



Standard Time Zones map

DAYCLI sample UTC / RDT

For example, a station that measures and stores its daily precipitation amount with the Reference Date and Time of UTC+8.
Its local practice is to measure the precipitation every day at 06:00 RDT and to assign this precipitation amount to the previous day.

With RDT

Station	Date (RDT)	RR
Station A	01/05/2025	0.0 mm
Station A	**02/05/2025**	**1.5 mm**
Station A	03/05/2025	5.0 mm
Station A	04/05/2025	0.0 mm
Station A	05/05/2025	2.8 mm
Station A	…	

With UTC

Station	Date (UTC)	RR
Station A	01/05/2025	1.5 mm
Station A	**02/05/2025**	**5.0 mm**
Station A	03/05/2025	0.0 mm
Station A	04/05/2025	2.8 mm
Station A	…	
Station A	…	

For example, for the day 02/05/2025 RDT it is assigned 1.5 mm of precipitation from the 02/05/2025 at 06:00 RDT to the 03/05/2025 at 06:00 RDT.

If we switch to UTC that gives:
from the 01/05/2025 at 22:00 UTC (RDT-8) to the 02/05/2025 at 22:00 UTC.

WMO OMM

Values are identical but with a mismatch of 1 day

Benefits to keep national practices

- ✓ Avoid using datasets with different conventions (national, regional or global).
- ✓ Avoid error-prone manipulation of dates and times.
- ✓ Be in accordance with current datasets providers (e.g. NOAA, BoM, Météo-France).
- ✓ Support and preserve current national practices.
- ✓ Simplify message preparation and post-processing.
- ✓ Enhance clarity and reduce the risk of time-based discrepancies.

Note:

- ✓ In climatology it is important to consider long-term data series in the most homogeneous way possible.
- ✓ DAYCLI is not managing Daylight saving time (DST)
- ✓ Most NMHSs use Reference Date and Time (RDT) because it reflects the conditions on their local days.
- ✓ Some stations do not have the hourly values that correspond to their daily values, meaning it is impossible to compute their daily data with other conventions.

Example of conventions for 3 daily datasets providers

- ✓ **NOAA/NCEI** and in particular its GHCND (**Global Historical Climatology Network Daily**), exchanges its daily data according to the time of observation that "*is the 24 hour clock time of the observation given as the local time at the station of record*". https://www.ncdc.noaa.gov/cdo-web/datasets and see https://www.ncei.noaa.gov/pub/data/cdo/documentation/GHCND_documentation.pdf

- ✓ **BoM** (Australia) exchanges its daily data according its Reference Date and Time (RDT). E.g. for rainfall at the station of Melbourne Airport , measurements are nominally made at 9 am local clock time and record the total for the previous 24 hours http://www.bom.gov.au/jsp/ncc/cdio/weatherData/av?p_nccObsCode=136&p_display_type=dailyDataFile&p_startYear=&p_c=&p_stn_num=086282

- ✓ **Météo-France** exchanges its daily data according the Reference Date and Time zones. That means in UTC for its territory in continental Europe (0 meridian), and in RDT for stations in Polynesia, in the Caribbean, etc. https://www.data.gouv.fr/fr/datasets/donnees-climatologiques-de-base-quotidiennes/

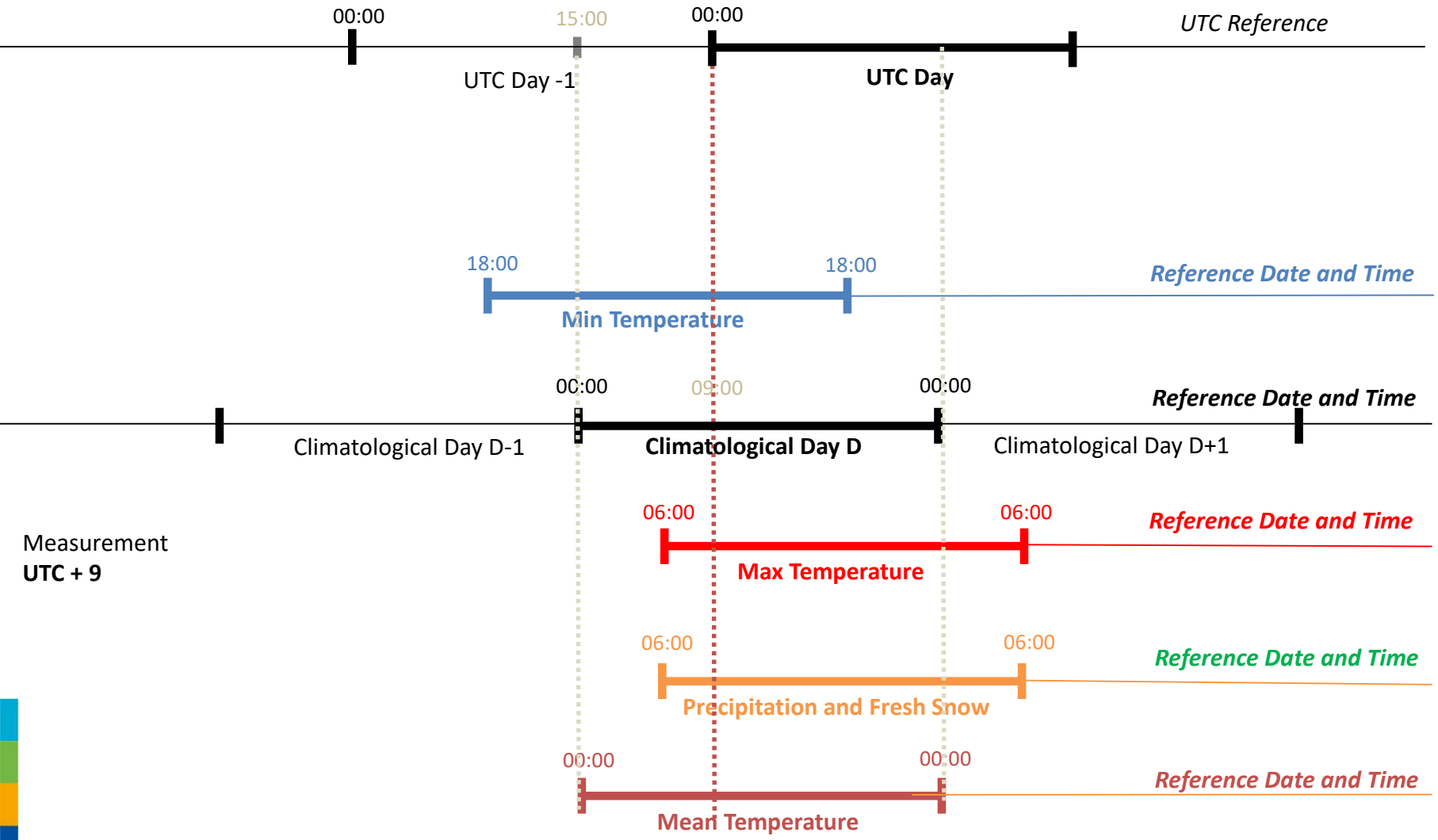
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Thank you !

WMO OMM

World Meteorological Organization
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DAYCLI: climatological practices sample



Measurement
UTC + 9

WMO OMM

DAYCLI history

Before 2012: Attempts to use SYNOP data for collection of daily parameters. This solution was abandoned due to difficulties on climatological practices around the world

2012: The Global Climate Observing System (GCOS) Atmospheric Observation Panel for Climate recommends to share daily observations values

2015: WMO approves the DAYCLI BUFR template **3 07 074**

2019-2020: Trial phase for the DAYCLI. It reveals issues and the need to modify the DAYCLI BUFR Template

2020: TT-TDCF & ET-DRC & WMO Members are collaborating to define the new DAYCLI BUFR Template

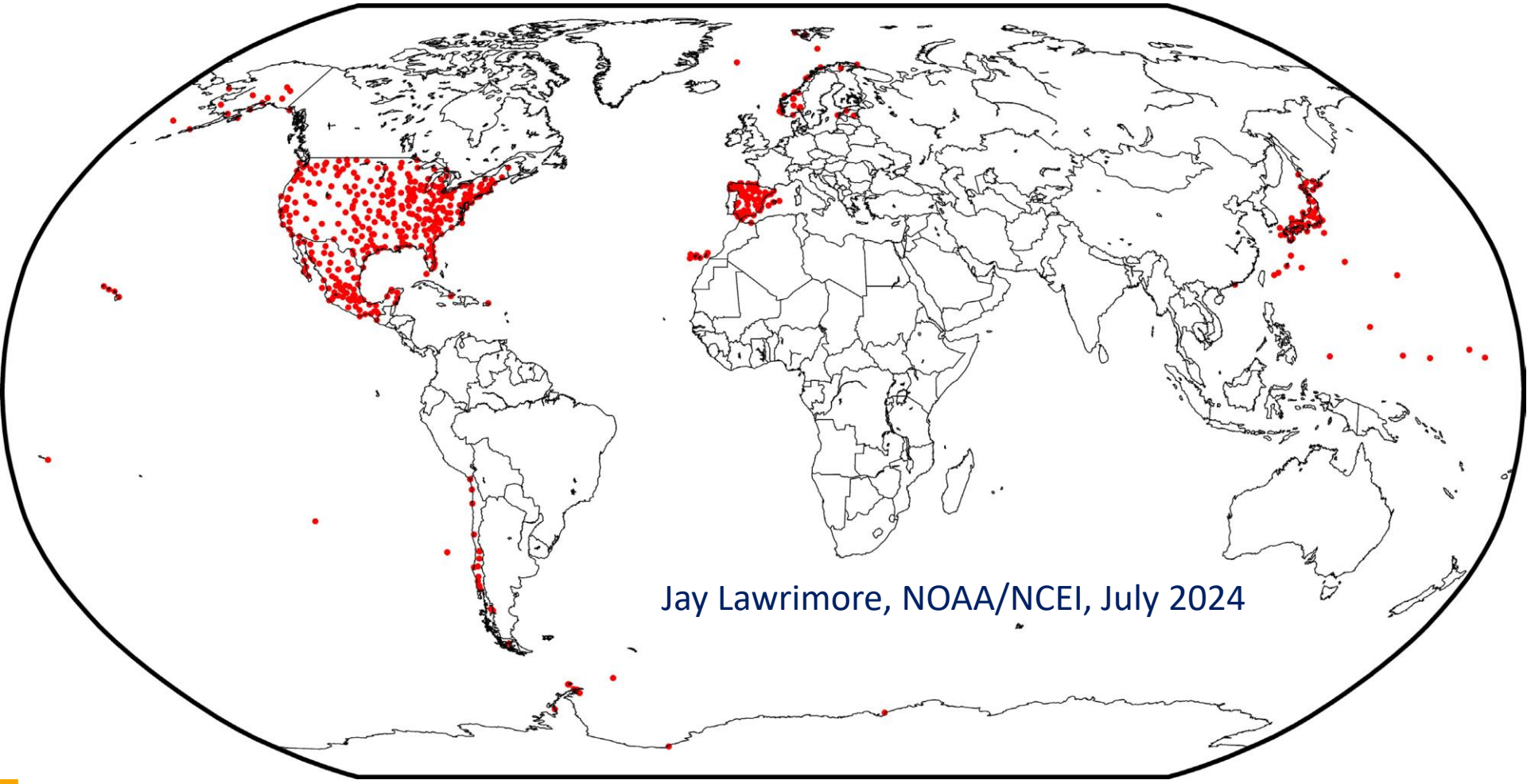
2022: BUFR Template for the DAYCLI message **3 07 075**, that replaces the 3 07 074, is adopted the 13 April, and reported the 15 May into the WMO Manual on Codes. https://github.com/wmo-im/BUFR4/issues/51

2024: Update of the WIGOS Manual (WMO-No. 1160) where Members should provide DAYCLI report

2024: Experts review that recommend modifications to preserve national and international practices for datasets integrity

As far as we know, no one is using the BUFR sequence 3 07 075 yet. Therefore, if it is validated, it is important to amend it in the very near future to avoid too much implementation work for NMHSs.

Countries Transmitting DAYCLI and Received at NCEI in 2024



So far, **no evaluation in the real world has been made** with the new DAYCLI BUFR template (3 07 075) through the WIS (to be verified today)